

Utilizing Internet Search Trends and Wastewater Surveillance to Identify Infectious Disease Outbreaks in Communities

Alessandro Zulli,^{||} Zoe Zhang,^{||} Madelena Ruedaflores, Jordan Sahly, Darryl Angel, Karthik Rohatgi, Waleed Malik, Ritche Hao, James Shepherd, and Jordan Peccia*



Cite This: *Environ. Sci. Technol.* 2025, 59, 3401–3410



Read Online

ACCESS |



Metrics & More



Article Recommendations



Supporting Information

ABSTRACT: This study proposes a novel approach for viral infectious disease surveillance using Google Trends data to model wastewater virus concentrations, providing a rapid, low-cost method for indicating outbreaks. Google Trends search terms were found to correlate strongly with wastewater viral concentrations and clinical cases for influenza A and respiratory syncytial virus ($R^2 = 0.76$ and 0.66). For norovirus and mpox, for which clinical data were limited, Google Trends showed significant correlations with wastewater concentrations. Three modeling approaches were developed: simple linear, stepwise selection, and principal component analysis. These models demonstrated strong predictive power for both norovirus (R^2 of up to 0.66) and mpox (R^2 of up to 0.60) wastewater concentrations. The approach was validated using a case study of a documented 2021 norovirus outbreak in Hartford, CT, where Google Trends indicators rose in tandem with wastewater concentrations, potentially providing earlier outbreak detection than clinical case data. This method offers a complementary data stream to wastewater surveillance for public health decision-making, particularly valuable in areas lacking a robust clinical testing infrastructure. Limitations include potential confounding factors, such as media coverage and the need to consider local idioms in international applications.

KEYWORDS: wastewater, Internet search trends, modeling, respiratory illness, infectious diseases



INTRODUCTION

The risk of contracting a viral infection within a community is strongly related to the percentage of the population that is currently infected, increasing substantially during periods of outbreaks. While clinical testing programs for viral infections have been used to track community infections, these programs are limited in scope, do not exist in many areas of the world, and outbreak data is rarely publicly available.^{1–3} The resulting lack of epidemiological information hampers the ability of citizens and public health officials to make rational decisions and implement nonpharmaceutical interventions that can mitigate or prevent infectious disease epidemics.

Viral outbreaks of Norovirus GI/Norovirus GII and Mpox Clade II exemplify this lack of outbreak information, even in locations with robust clinical testing infrastructure. In the U.S., norovirus outbreaks have significant and often undiagnosed effects in tertiary care hospitals and nursing homes, with an estimated 21 million cases resulting in more than 100,000 hospitalizations.^{2,4–7} However, norovirus monitoring poses several challenges, with up to 30% of cases being asymptomatic and a majority of cases going undetected due to limited testing regimens coupled with low disease severity.^{1,7,8} Mpox Clade II is an emerging DNA virus that led to several outbreaks across the United States in 2022. Over 30,000 cases were detected in

the United States during this period, though a significant percentage went undiagnosed in high-risk and vulnerable populations.^{9,10} In these cases and others, wastewater viral surveillance has been shown to be an effective way to track a variety of viral pathogen outbreaks, and it has been demonstrated that the time course of wastewater virus concentrations closely correlate with reported clinical cases.^{11–14} While wastewater surveillance has been used to track both norovirus and mpox and has demonstrated some consistency with the limited clinical case data available,^{13,15} wastewater surveillance is still a resource-intensive endeavor, requiring the type of capital investments and technical expertise that are not available in many parts of the world.

Google Trends data has previously been used to predict outbreaks of influenza in the United States.¹⁶ These systems have shown to be low cost and easy to implement but suffered from several issues when trained using clinical case data.^{17–19}

Received: September 16, 2024

Revised: February 3, 2025

Accepted: February 5, 2025

Published: February 12, 2025



The systems are often sensitive to media biases, where coverage of an illness can significantly impact predictions, leading to overestimations when the model is not updated, such as the case of Google Flu Trends.^{20,21} Wastewater-based epidemiology can mitigate these issues, allowing us to train Google Trends-based models on an unbiased, highly time-resolved ground truth, allowing for adjustments to be made rapidly and for the detection of any anomalous predictions. This work focuses on norovirus, which is specifically less susceptible to these biases as it is common and rarely fatal and does not often see increased media attention to the same degree as viruses such as mpox and avian influenza.

Here, we propose an approach for viral infectious disease surveillance where widely available Internet Google Trends searches are validated by wastewater surveillance and utilized in models that can indicate infectious disease outbreaks in a community. There is mounting evidence that Google Trends searches reflect the underlying epidemiology of several infectious diseases such as influenza, measles, dengue, and Zika fever.^{17,22–29} In this study, we use Google Trends data to build predictive models for wastewater concentrations that can better reflect the underlying epidemiological situation than clinical case data. We demonstrate a specific case application of this methodology to a norovirus outbreak in Hartford, CT, in 2021. This rapid, calibrated, low-cost approach can be an important tool for indicating infectious disease outbreaks, especially in locations where limited clinical or wastewater surveillance exists.

METHODS

Wastewater Samples. This study covered from July 1, 2022 to May 1, 2023 in New Haven, CT, and spanned the cold and flu season in the northeastern U.S. Outbreaks of Human orthopneumovirus (RSV A/B), influenza A virus, norovirus, and mpox occurred during this period. Primary sewage sludge samples 45 mL in volume were collected daily from the East Shore Water Pollution Abatement Facility. Primary sludge was collected as viruses have been shown to strongly partition into wastewater solids.^{30–32} This facility serves a population of approximately 200,000 including the municipalities of New Haven, Hamden, East Haven, and parts of Woodbridge, CT. Figure S3 is a sewer service map of this plant and demonstrates the coverage of these cities. All samples were collected between 8:00 a.m. and 10:00 a.m. and transported to Yale University laboratories on ice. Grab samples of primary sludge were collected once they had passed through primary sedimentation and gravity thickening.^{11,12} To test our model, we also used primary sludge wastewater samples from the Hartford, CT South Meadows wastewater treatment plant. These Hartford samples were collected daily from April 22, 2021 to June 10, 2021.

Influenza A, RSV A/B, Norovirus GI/GII, and mpox Viral RNA Concentration Quantification. Total nucleic acids were extracted from primary sludge samples using an automated extraction process on the Kingfisher Apex system (ThermoFisher Scientific #5400940). A MagMAX Wastewater Ultra Nucleic Acid Isolation Kit was used to process 2 mL of sample with the addition of 400 μ L of phenol-chloroform-isoamyl alcohol (25:24:1) during the initial lysis step (ThermoFisher Scientific #A52606, ThermoFisher Scientific #15593031). Samples were attached to an orbital shaker and mixed at top speed (250 rpm) for 15 min for thorough lysis. Samples were then spun down at 3000 g for 15 min. A total of

1.96 mL of spun-down supernatant was used as initial input and eluted into 200 μ L of the kit-provided elution solution. A more thorough extraction protocol is provided in the Supporting Information. Total nucleic acid concentrations were quantified by spectrophotometry (NanoDrop 2000, ThermoFisher Scientific). Droplet digital PCR assays were taken from peer-reviewed literature and Bio-Rad Laboratories (Bio-Rad dEXD8484464, Bio-Rad dEXD77482599).^{11,33,34} All assays were validated internally by using gBlock gene fragments ordered from Integrated DNA technologies as positive controls. Droplet digital PCR (ddPCR) concentrations were normalized using the total nucleic acid concentrations as previously described.¹² Assay details, primer/probe concentrations, and thermocycling conditions are provided in the Supporting Information. All positive and negative controls performed as expected.

Epidemiological and Google Trends Data. Clinical case data were collected from the Yale New Haven Hospital (YNHH) electronic records system. Case data were provided by the date of specimen collection and confined to the hospitals and clinics operating in the wastewater catchment area. Pearson correlation coefficients and their corresponding significance were determined to assess the strength of linear correlation between wastewater virus concentrations and clinical case signals. These analyses were run using Rstudio with the R packages stats v.4.3.0 and PerformanceAnalytics v.2.0.4. Clinical case data related to the Hartford norovirus outbreak was based on the date of specimen collection and was collected from the Connecticut Department of Health report on food-borne outbreaks.³⁵

Google Trends data is updated in real-time and was collected directly from the Google Trends Web site. The geographic location was set to New Haven, CT, and the date range was set to January 1, 2022 to July 1, 2023 for all search terms. When possible, appropriate search term subsets were used (i.e., Ailment – “vomiting” rather than Search term – “vomiting”). Subsets represent terms related to the overall term, such as including “vomit”, “vomiting”, and “throwing up” under the category of Ailment – “vomiting”. Google Trends data for influenza A virus and RSV A/B were collected based on their common names, “influenza” and “RSV”. Norovirus search data were collected based on its common name, “norovirus”, and symptoms clinically associated with infection: “gastroenteritis”, “vomiting”, and “diarrhea”. Mpox search data were collected based on its common name and associated search terms. For each virus, we identified and tried a list of search terms related to these categories (virus, symptom, illness/diagnosis, treatment/medication, vaccine, exposure) to reflect a broad sense of the public’s perception of potential viral outbreak and infection (see Table S6 for the list of search terms). For each Internet search term, Google Trends provides a search volume index ranging from 0 to 100, which measures search intensity/popularity (number of searches for a given keyword divided by total search volume at a point in time) by geography and time range.³⁶

We used the *trendecon* open-source R package to extract data from Google Trends and to construct high-frequency and time-consistent daily time series, where search interest over time is comparable at different (and/or distant) points in the time of the study.^{37,38} Google Trends does not allow for daily data over periods longer than 90 days. Thus, the Google Trends API was called multiple times at daily, weekly, and monthly frequencies and harmonized to preserve the long-term

trend while maintaining daily data resolution for our analysis. The constructed daily time series may exceed the original 0 to 100 Google Trend index range due to interpolation and temporal disaggregation that was performed in the R package *tredecon*.

Statistical Analyses. Norovirus wastewater concentrations were measured separately for norovirus GI and norovirus GII. These two strains are responsible for the majority of norovirus outbreaks and illnesses worldwide.^{1,7} The two strains have the same symptomatic presentation and require targeted PCR testing to distinguish.⁸ For the purposes of this study, a weighted composite measure of the two concentrations was calculated and used for comparisons to clinical testing and Google Trends, which do not distinguish strains. Specifically, norovirus GI and GII concentrations are rescaled using a min-max normalization to the range of 0 to 100. The two converted concentrations were given equal weight in the composite wastewater signal by performing a simple average. We also standardized clinical testing data so that wastewater, cases, and Google Trends are all on the same 0–100 scale. We used the *rescale* function from the *scales* R package to perform a linear min-max transformation of all variables, adjusting their values proportionally to a 0 to 100 scale.

We used two approaches to build relationships between the wastewater viral RNA/DNA concentrations and Google Trends. These include a principal components analysis (PCA) approach and a multivariable statistical model approach. PCA is a method of linear dimensionality reduction and feature extraction that retains most of the variation, information, and signal in the original search interest data. We use this technique to decrease the number of search term indicators for consideration. The new uncorrelated variables - principal components (PCs) were constructed using unprocessed data through a linear combination of the original search terms variables. R packages *FactoMineR* v2.8 and *factoextra* v1.0.7 were used to visualize the result of PCA. The PCA models were inspected through PCA scores and loading graphs, and the PCA scores were subsequently used in time-series modeling. The amount of explained variance (eigenvalues) per newly defined PC was inspected with scree plots, where larger eigenvalues correspond to more variance being explained by the data. We retained all components before the elbow point on a scree-plot. This corresponds to the first four PCs for norovirus time-series modeling and first three PCs for mpox modeling.

In addition to the PCA approach, simple and multivariate linear regression models were constructed between Google Trends search terms and wastewater concentrations. The model structure is as below. We performed bidirectional stepwise model selection using Akaike Information Criteria (AIC) to determine the best subset of the predictors. We further assessed the models through coverage probability by calculating prediction intervals (calculated as the proportion of times that measured RNA concentrations fell within the 95% prediction interval of the model-estimated RNA concentrations). StepAIC from the MASS (7.3-60) R package was used. This selection process starts with no variables in the model and builds a regression model by successively adding or removing variables based on the *t*-statistics.

$$\text{wastewater}_t = \beta_0 + \sum_{i=1}^n \beta_i x_{i,t} + \epsilon_t$$

where:

wastewater_t is the predicted wastewater concentration at time t .

β_0 : intercept of the regression model.

β_i : regression coefficient.

$x_{i,t}$: the value of the i th Google Trends search term or the i th principal component at time t , obtained through PCA of the original search term predictors.

ϵ_t : the residual error at time t , representing the portion of wastewater_{*t*} not explained by the model.

RESULTS

Google Trends Search Results Correlate with Both Wastewater Surveillance Data and Clinical Cases for Major Respiratory Viruses. Influenza A and RSV are two clinically relevant respiratory viruses that have significant human health impacts, with known seasonal patterns and robust clinical testing programs. To determine the potential for Google Trends as an indicator of underlying infections, we compared Google Trends search terms for the common name of the two viruses influenza and RSV to wastewater concentrations and clinical case rates on a daily basis. Figure 1 shows the results of these preliminary analyses, plotting the three signals together.

Comparison of Norovirus and mpox Cases and Wastewater Surveillance Data. To extend the use of

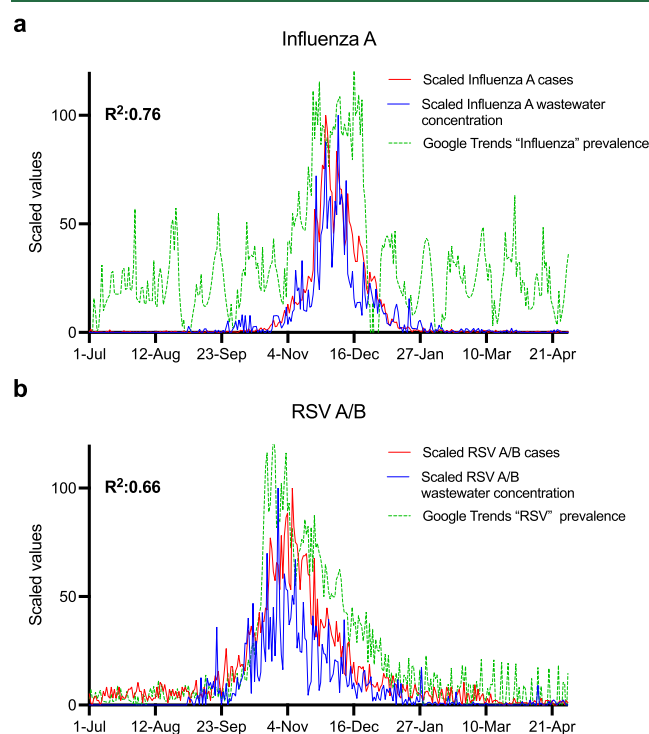


Figure 1. Simple temporal comparisons of three outbreak signals: Google Trends data, clinical cases, and wastewater virus RNA concentrations, July 1, 2022 to May 1, 2023. (A) Shows a comparison between the search term influenza, daily clinical influenza cases as measured by Yale New Haven Hospital System electronic health records, and daily wastewater RNA concentrations of Influenza A virus. (B) Shows a similar comparison for the search term RSV, daily clinical RSV cases as measured by Yale New Haven Hospital, and daily wastewater RNA concentrations of RSV A/B virus. Adjusted correlation coefficients between wastewater concentrations and Google Trends data are shown.

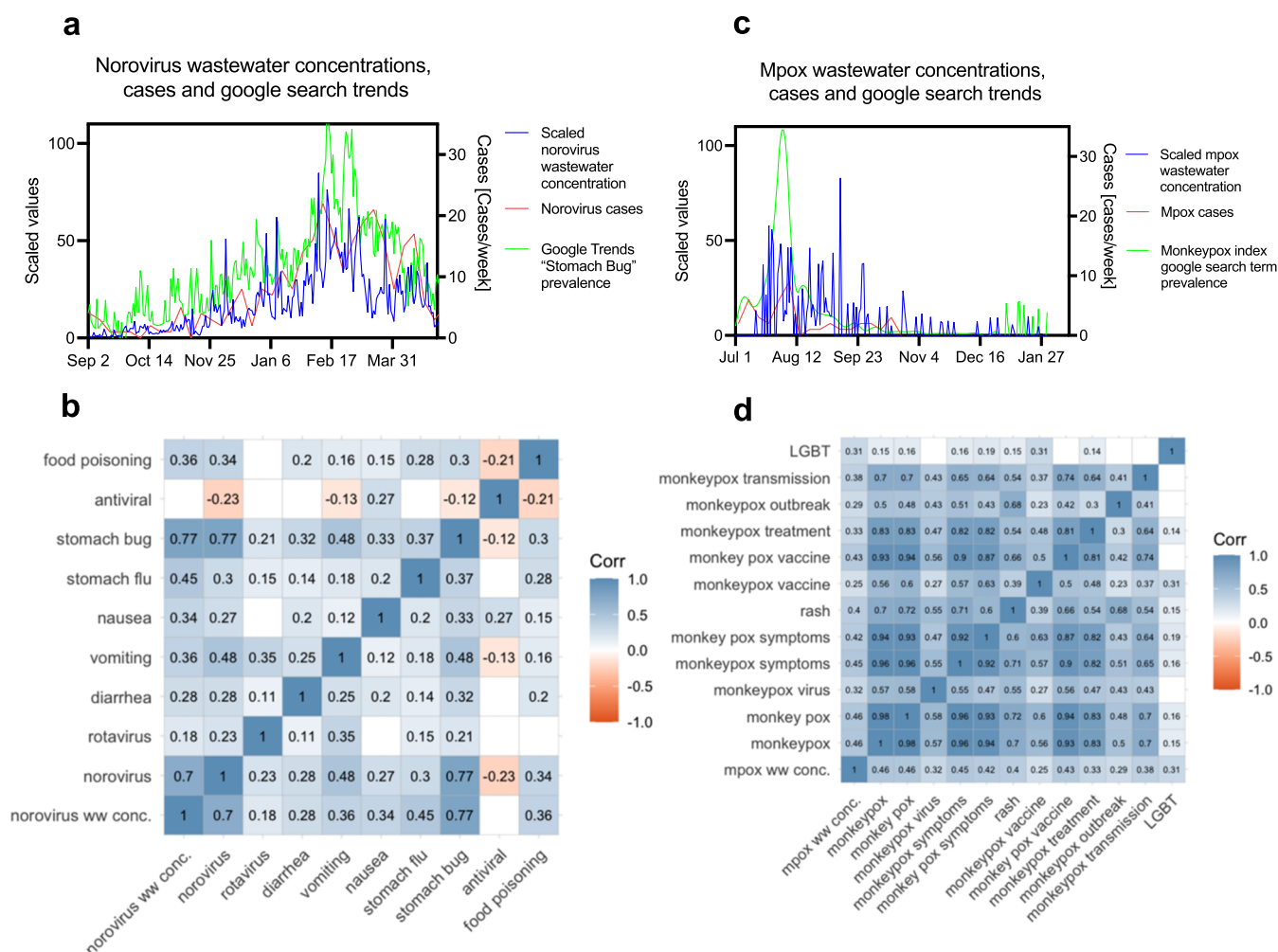


Figure 2. Relationships between individual Google Trends search terms, wastewater pathogen concentrations, and clinical case information. (A) Shows a time-series comparison between daily wastewater concentrations of norovirus, weekly clinical case data, and daily Google Trends indicators. The correlation between weekly wastewater concentrations of norovirus and weekly clinical case data was 0.80 ($p < 0.05$). (B) Shows a correlation matrix (Pearson correlation) of all relevant Google Trends search terms, norovirus wastewater concentrations and clinical cases. Norovirus ww conc. represents norovirus wastewater concentrations. (C) Demonstrates a time-series comparison between daily wastewater concentrations of mpox, weekly clinical mpox data, and daily Google Trends search terms. The Correlation between weekly wastewater concentrations of mpox and weekly clinical case data was 0.47 ($p < 0.05$). (D) Shows a correlation matrix (Pearson correlation) of all relevant Google Trends search terms, mpox wastewater concentrations, and clinical cases. Mpox ww conc. represents mpox wastewater concentrations. Both “monkeypox” and “monkey pox” were used in search term analysis as both have been commonly used in Google searches. Search terms that were not significantly correlated with wastewater concentrations were removed. The shading in panels (C) and (D) represents the strength and direction of correlation, with darker blue indicating stronger positive correlations, darker red indicating strong negative correlations, and white indicating correlations that are statistically insignificant ($p < 0.05$).

Google Trends indicators for norovirus and mpox, we ran simple regression analyses to compare clinical case information, wastewater concentrations, and Google Trends search terms (Figure 2). Figure 2A shows a time-series presentation of our norovirus wastewater concentrations, norovirus cases as recorded by Yale New Haven Health’s electronic medical records system, and the Google Trends search term “stomach bug”. Notably, Google Trends data match wastewater concentration dynamics closely, although clinical cases reflected a very low case burden. Figure 2C shows a correlation matrix of all Google Trend search terms tested with significant correlations ($p < 0.05$). The strongest among these was a correlation between searches for the term “stomach bug” and norovirus wastewater concentrations at an R^2 of 0.77 ($p < 0.001$). “Stomach bug” is an informal term, which typically refers to symptoms matching a clinical diagnosis of gastro-

enteritis. We observed significant correlations between Norovirus wastewater concentrations and terms typically associated with norovirus, including “diarrhea”, “vomiting”, and “norovirus”.

For mpox, the same correlation analyses were performed. Figure 2B shows a time series of mpox wastewater concentrations in New Haven, mpox cases as recorded by Yale New Haven Health’s electronic medical records system, and a selection of individual Google Trends indicators. Figure 2D shows a correlation matrix of all terms with significant correlations ($p < 0.05$). The strongest among these was searches for “monkeypox” with an R^2 of 0.46 ($p < 0.001$). Due to limited case data, wastewater concentrations could represent a better indicator of the underlying epidemiological situation and that Google Trends search terms could be used for further statistical analysis and modeling.

Google Trends-Based Predictive Models. Based on the results of our preliminary analysis, we modeled norovirus and mpox virus wastewater concentrations as a function of Google Trends search terms. Three approaches are presented to demonstrate their applicability in different contexts. First, due to the strong correlations between Google Trend terms and norovirus wastewater concentrations, we created a simple linear model for norovirus that used a single variable, “stomach bug”, as our explanatory variable and a scaled concentration of norovirus as the response variable. Figure 3A shows the results of this modeling, with the predicted concentrations closely matching observed values (R^2 : 0.59, $p < 0.001$). For the second model, we used a bidirectional stepwise selection model that started with no variables (i.e., Google Trend search terms) and tested the addition of each variable. The final model included 7 variables: norovirus, nausea, stomach flu, stomach bug,

vomiting, rotavirus, and diarrhea. Figure 3B shows that this model has a similar performance to the single variable model (Adjusted R^2 : 0.66, $p < 0.001$). Lastly, we tested a model that used principal component analysis to reduce the number of explanatory variables. Figure 3C demonstrates this model's performance when using the first 2 principal components (Adjusted R^2 : 0.59, $p < 0.001$). Predictions intervals generated from these models performed as expected (94%, 96%, 95%, respectively), with ~95% of values falling within the prediction interval.

We then performed similar modeling for the MPox data, again seeking to model wastewater concentrations as a function of the Google Trends searches. Preliminary analysis of mpox data did not show any single dominant Google Trends predictor (Figure 2D). We therefore created two models, one using the stepwise selection process detailed above and the other using principal component analysis (Figure 4). Both the

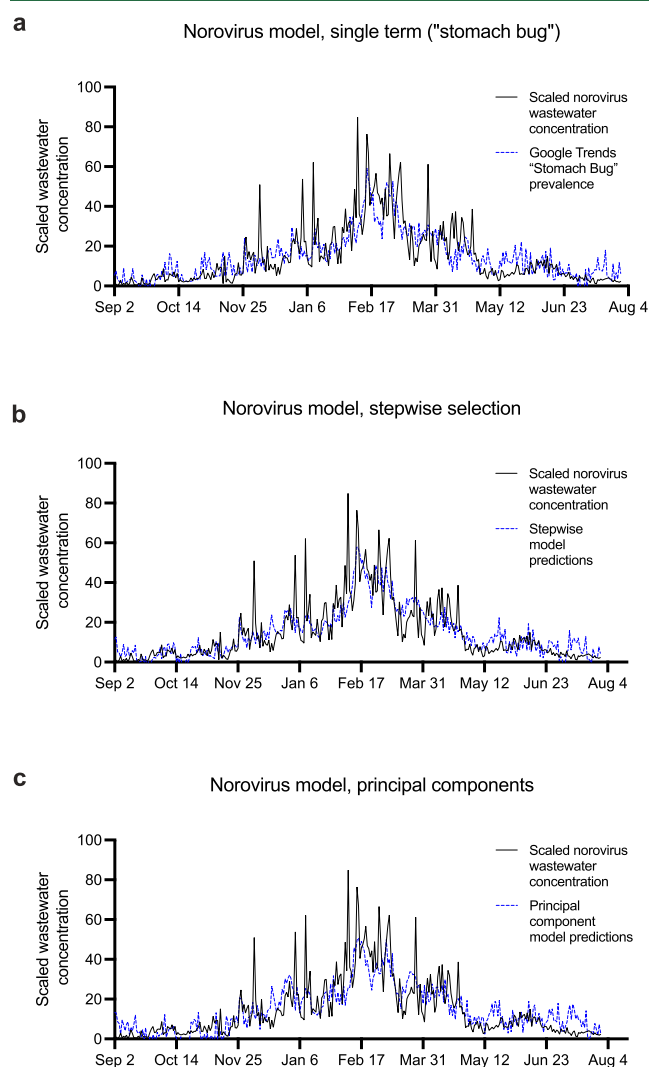


Figure 3. Modeling approaches for utilizing Google Trends data to predict norovirus wastewater concentrations with predictions in blue and measured wastewater concentrations in black from September 2 to August 1. (A) Demonstrates a simple single variable model using a strongly correlated Google Trends search term, “stomach bug” (R^2 : 0.59). (B) Demonstrates the results of a model trained using stepwise selection against measured wastewater concentrations (R^2 : 0.66). (C) Demonstrates a model trained using the first two principal components (R^2 : 0.59). The data are from 2022 to 2023.

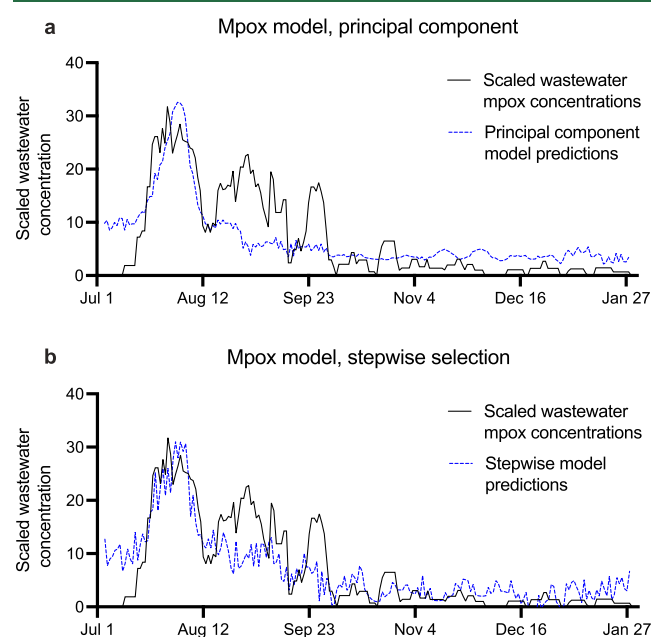


Figure 4. Multiple modeling approaches for utilizing Google Trends data to predict mpox wastewater concentrations with predictions in blue and measured wastewater concentrations in black. (A) Demonstrates a model trained using the first two principal components (Adjusted R^2 : 0.44, $p < 0.001$). (B) Demonstrates the results of a model trained using stepwise selection, against measured wastewater concentrations (Adjusted R^2 : 0.60, $p < 0.001$). The data are from 2022 to 2023.

principal component model and stepwise selection model match observed concentrations during the initial outbreak (Adjusted R^2 : 0.44, 0.60, $p < 0.001$), though they missed peaks in August and September. This may be due to a potential initial media bias in Google Search trends. Prediction intervals generated from these models performed as expected, though intervals were larger than for our norovirus modeling (96 and 94%, respectively).

Hartford Case Study. We applied the Google Trends modeling approaches described above to identify a prior food-borne outbreak of norovirus GII centered on a donut shop near Hartford, CT, in May 2021. Here, the Google Trends-based models (PCA model and stepwise model) identify this acute outbreak event ahead of the Connecticut Department of

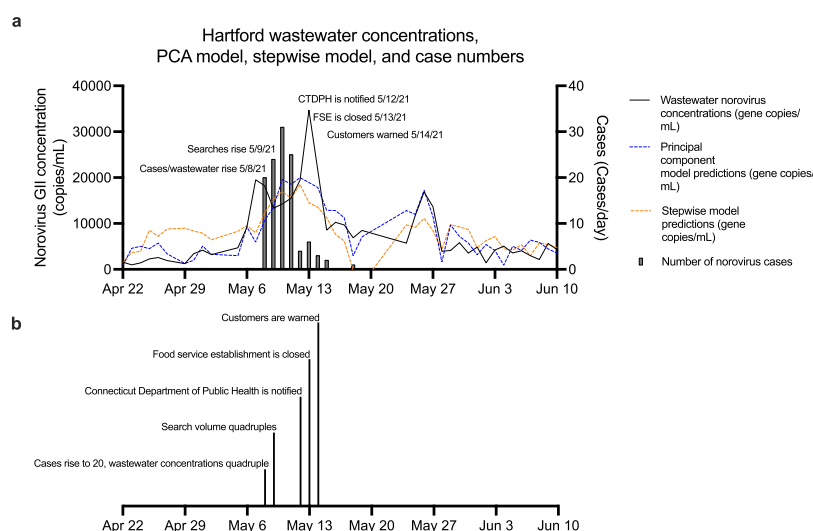


Figure 5. (A) is a graphical comparison of norovirus GII concentrations in Hartford wastewater with Google Trends-trained prediction models and daily case data. The graph shows wastewater concentrations (solid line), predictions from our principal component-based modeling, concentration predictions from our stepwise model (dotted lines), and case counts (bars). (B) Shows key outbreak events and their timing.

Health being notified of a potential outbreak (Figure 5), mainly based on the “stomach bug” search term. A retrospective investigation of reported cases by date of specimen collection found that norovirus cases increased from 0 on May 7, 2021, to greater than 20 cases by May 11, 2021. The Connecticut department of health was notified about a possible outbreak on May 12, 2021, and issued a warning on May 14, 2021. By comparison, wastewater concentrations of norovirus GII in Hartford’s South Meadow wastewater treatment facility began rapidly increasing May 7, more than quadrupling by May 8. Search volumes for “gastroenteritis” and “stomach bug” saw similar increases by May 9. The application of our stepwise and principal component models built using the above New Haven, CT data resulted in strong agreement with wastewater trends in Hartford during the outbreak, even at times that preceded the Department of Health warnings.

DISCUSSION

General Observations. The purpose of this study was to determine the feasibility of utilizing Google Trends as an indicator of viral infection outbreaks in an underlying population. For diseases including RSV and influenza, where both wastewater data and cases are in strong agreement, we demonstrated that simple models that utilize locally constrained Google Trends search terms can provide an accurate reflection of outbreaks. By leveraging additional wastewater data, we then developed and tested three different Google Trends-based models for identifying outbreaks of viral disease of norovirus and mpox where testing is not commonly available or known to have significant limitations. While the usage of primary sludge may limit the applications to plants with primary clarifiers, primary sludge provides a more consistent signal from wastewater than raw sewage, better reflecting the underlying population dynamics. This approach could be expanded to raw wastewater, either by directly extracting the wastewater or by separating the solids and performing analysis on the solids.^{14,31,32} Together, our results demonstrate the viability of Google Trends as a rapid, low-cost indicator of infectious disease outbreaks and the value of viral

wastewater concentrations as a confirmatory signal of the underlying epidemiological situation.

Wastewater virus concentrations are an appropriate metric to calibrate Google Trends-based models. Previous explorations of Google Trends and other search query-based epidemic detection depended on clinical case data as a confirmatory signal, which was often sparse and subject to testing biases.^{3,22,23,27,29} Further, these previous attempts were often prone to overfitting due to the limited training data available. Using wastewater concentrations as the confirmatory signal allows for a significant increase in the utility of Google Trends-based models by expanding potential targets while maintaining population level accuracy. Where clinical case information is available, wastewater has been demonstrated to be a highly accurate indicator of clinical illness in a population for a wide variety of viral infections.^{11–14,39,40} In many cases, wastewater monitoring has shown to be a more accurate indicator of viral infections than clinical testing when accounting for the potential biases of testing, such as access, prevalence of take-home testing, and social biases.^{3,10,41} Despite its low cost and rapidity when compared to clinical case monitoring, wastewater surveillance still requires significant capital investments and a centralized wastewater treatment plant. Google Trends analysis for epidemiology provides communities with a near-zero cost epidemiological indicator that has predictive power comparable to that of wastewater signals.

Google Trends data has previously been used as a potential indicator of infection for influenza, with studies demonstrating significant correlations between descriptive search terms such as “H1N1”, influenza, and “fever” and total numbers of influenza-like illness.^{25,42} Similar studies have been performed for RSV, norovirus, and mpox that use Google Trends and other social media trends to assess public interest in these viruses, but no such studies have used wastewater as the predicted variable and validating signal. Current studies on RSV and search trends are largely limited to using weekly hospitalizations or clinical case data, which limits the ability to provide a leading signal, and is limited to validation and training in areas that have significant and robust testing systems.^{43–46} The limit of using weekly clinical data also necessarily limits the potential lead time of Google Trends data

and hampers its usage to enact timely interventions. The geographic overlap of these data sets was also often limited to the state level, which can further confound model training.^{43,44} Studies on norovirus and search trends ran into similar but more pronounced issues, being limited to acute outbreaks that were investigated and reflected in clinical data.^{18,47–49} Lastly, an analysis of search trends and clinical data for mpox revealed how biases in media and clinical testing significantly affect the utility of search trends when not grounded to an unbiased signal.^{50,51}

A major failure of previous attempts at implementing Google Trends-based outbreak detection, particularly in the case of Google Flu Trends, was the lack of an unbiased real-time signal to ground the system and a failure to account for changes in search behavior.^{20,21} Using wastewater data allows for both detection of when these models do not match the underlying epidemiological situation and for the recalibration of these models in a rapid manner when compared to clinical case data. This grounding of the models allows us to minimize the impact of “big data hubris”, as we present these models as an evolution and complement to existing and emerging surveillance systems.²¹ As mentioned above, this becomes even more valuable when applied to viruses like norovirus, for which very little case data is available and for which norovirus wastewater concentrations can truly provide a ground truth.

Overall, our approach demonstrates significant advantages, addressing these limitations in clinical data, removing many potential biases, and preventing overfitting of models by training on frequent wastewater data. Beyond the significant increase in targeted viruses, this study presents at least two significant advances in using Google Trends as an epidemiological indicator. First among these is the development of wastewater concentrations (not case data) as a validating signal, particularly in the case of norovirus and mpox, which have limited clinical data available. Second, we strengthen this approach through the utilization of multivariate statistical models for norovirus and mpox, enacting variable selection through our wastewater indicators. For norovirus, we focused on terms that represented the clinical manifestation of the illness, as the name itself did not enter the general lexicon until outbreaks were well underway.^{52,53} Our strongest single search term correlation was found to be “stomach bug”. While this search term was significantly predictive in the region and study period considered, variable selection and the consideration of local idioms and dialects will play important roles in future implementations. We considered a variety of other search terms, many of which were strongly and significantly correlated to norovirus wastewater concentrations, such as “vomiting”, “food poisoning”, and “norovirus”. Multivariate, stepwise selected models and principal component-based models that used a variety of Google Trends terms demonstrated improved performance and strong fits with wastewater data for both norovirus and mpox. The approach described in this paper addresses some of the previous issues with using Google Trends for epidemiological prediction, such as overfitting, by providing a continuously updated unbiased indicator of infection to train against and significantly increasing the size and fidelity of the training data set.^{21,54}

Wastewater surveillance is an effective, low-cost indicator of infection in human populations and has become an important data stream for public health.^{14,39} By the development of these Google Trends models, another data stream has become available for public health decision-makers to use in

conjunction with other surveillance systems. Google Trends predictions could allow for confirmation of wastewater signals ahead of clinical case signals and allow for a more rapid response time in the case of outbreaks. This is clearly demonstrated by the timeline of events in the norovirus outbreak in Hartford, where Google Trends increased 3 days before public health departments were notified of rising clinical cases.

Case Study. To further validate our Google Trends approach to outbreak predictions, we analyzed a prior norovirus GII outbreak in Hartford, Connecticut, in the spring of 2021. The results of applying our calibrated models to this outbreak demonstrated that the rises in Google Trends indicators rose in tandem with cases and wastewater concentrations, and our stepwise selection regression model closely matched the wastewater norovirus GII concentrations.

Search terms related to the outbreak increased 4-fold a day after significant increases in wastewater concentrations and a full 3 days before the CTDPH was notified of a potential outbreak of norovirus. Google Trends increased significantly approximately 1 day after wastewater concentrations. By comparison, preliminary case data were not available until several days after the initial infections and were not completely assembled until several months later in retrospective outbreak investigations. Google Trends indicators would provide a real-time, low-cost indicator for surveillance purposes while maintaining a similar degree of accuracy to wastewater-based surveillance. This is reflected by the notable agreement between the term “stomach bug” and cases (R^2 : 0.63) and between our stepwise prediction model and wastewater concentrations (R^2 : 0.57). The Google Trends model could have provided faster notification to public health authorities of the outbreak than clinical case data, allowing earlier implementation of nonpharmaceutical interventions to prevent further community spread of disease. Significant additional work is necessary toward the expansion, refinement, and validation of Google Trends as an indicator for infectious diseases, particularly in the case of diseases with limited clinical surveillance. This validation gap is potentially addressed by environmental surveillance such as wastewater-based epidemiology, allowing the models to be validated for diseases that lack expansive clinical testing.

Limitations. These methods have limitations that must be considered in their application and development. Certain search terms, such as influenza, appear to have a nonzero baseline level of popularity which needs to be investigated and potentially corrected for. Confounding factors such as coverage in local and national media and communications from health agencies must be considered in future implementations and studies. The main consideration is the effect of public awareness of terms on Google Trends. News articles and viral social media content could significantly affect the timing and volume of searches for particular terms.¹⁷ As an example, searches for the term “norovirus”, when compared to searches for the more generic term “stomach bug”, were significantly impacted by articles published about outbreaks in the northeastern U.S. in January–February 2023.⁵² For mpox, the initial media coverage of the outbreaks and significant public concern led to a much higher initial search volume during the early phase of the outbreaks as compared to the downtrend. While this bias is likely present for other emerging viruses such as avian influenza, norovirus, a much more common pathogen, did not demonstrate any such media biases. Further, there

must be additional analysis of the trustworthiness and stability of published Google Trends data, which is provided by a private for-profit entity rather than a regulated government agency. Another limitation is that when applying these methods internationally, efforts must take into consideration local idioms rather than direct translations of terms. While Internet access continues to grow worldwide, it is far from ubiquitous in lower-income countries. Lastly, the determination of search terms to include in the initial analyses requires significant research and collaboration but is necessary to avoid the biases introduced by algorithmic approaches, which lead to overfitting.^{21,54}

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.4c05723>.

Molecular assay details, PCA details, extraction details, sewer service map (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Jordan Peccia – Department of Chemical and Environmental Engineering, Yale University, New Haven, Connecticut 06511, United States; orcid.org/0000-0002-6482-2084; Email: Jordan.Peccia@Yale.edu

Authors

Alessandro Zulli – Department of Chemical and Environmental Engineering, Yale University, New Haven, Connecticut 06511, United States; orcid.org/0000-0003-3636-0211

Zoe Zhang – Department of Chemical and Environmental Engineering, Yale University, New Haven, Connecticut 06511, United States

Madelena Ruedaflares – Department of Chemical and Environmental Engineering, Yale University, New Haven, Connecticut 06511, United States

Jordan Sahly – Department of Chemical and Environmental Engineering, Yale University, New Haven, Connecticut 06511, United States

Darryl Angel – Department of Chemical and Environmental Engineering, Yale University, New Haven, Connecticut 06511, United States

Karthik Rohatgi – School of Medicine, Yale University, New Haven, Connecticut 06511, United States

Waleed Malik – School of Medicine, NYU, New York City, New York 10016, United States

Ritche Hao – School of Medicine, Yale University, New Haven, Connecticut 06511, United States

James Shepherd – School of Medicine, Yale University, New Haven, Connecticut 06511, United States

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acs.est.4c05723>

Author Contributions

[†]A.Z. and Z.Z. contributed equally to this work.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was funded by a gift from Jonathan Rothberg's 4Catalyzer to Yale University. We thank the Water Pollution Control Authority in the City of New Haven, CT, USA, for assistance with daily wastewater sampling.

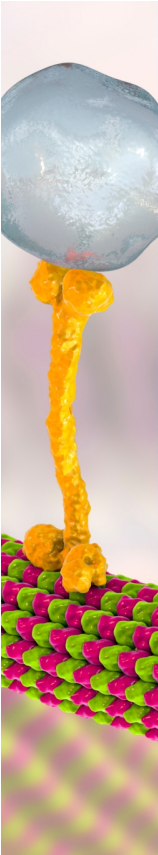
■ REFERENCES

- (1) Miura, F.; Matsuyama, R.; Nishiura, H. Estimating the Asymptomatic Ratio of Norovirus Infection During Foodborne Outbreaks With Laboratory Testing in Japan. *J. Epidemiol.* **2018**, *28* (9), 382–387.
- (2) Beersma, M. F. C.; Sukhrie, F. H. A.; Bogerman, J.; Verhoef, L.; Mde Melo, M.; Vonk, A. G.; Koopmans, M. Unrecognized Norovirus Infections in Health Care Institutions and Their Clinical Impact. *J. Clin. Microbiol.* **2012**, *50* (9), 3040–3045.
- (3) Boehm, A. B.; Wolfe, M. K.; White, B.; Hughes, B.; Duong, D. Divergence of Wastewater SARS-CoV-2 and Reported Laboratory-Confirmed COVID-19 Incident Case Data Coincident with Wide-Spread Availability of at-Home COVID-19 Antigen Tests. *PeerJ* **2023**, *11*, No. e15631.
- (4) Norovirus Worldwide | CDC, 2023. <https://www.cdc.gov/norovirus/trends-outbreaks/worldwide.html>.
- (5) Harris, J. P.; Iturriza-Gomara, M.; O'Brien, S. J. Re-Assessing the Total Burden of Norovirus Circulating in the United Kingdom Population. *Vaccine* **2017**, *35* (6), 853–855.
- (6) Forman, D.; de Martel, C.; Lacey, C. J.; Soerjomataram, I.; Lortet-Tieulent, J.; Bruni, L.; Vignat, J.; Ferlay, J.; Bray, F.; Plummer, M.; Franceschi, S. Global Burden of Human Papillomavirus and Related Diseases. *Vaccine* **2012**, *30*, F12–F23.
- (7) Norovirus Burden and Trends | CDC, 2023. <https://www.cdc.gov/norovirus/burden.html>.
- (8) Robilotti, E.; Deresinski, S.; Pinsky, B. A. Norovirus. *Clin. Microbiol. Rev.* **2015**, *28* (1), 134–164.
- (9) 2022 U.S. Map & Case Count | Mpox | Poxvirus | CDC, 2023. <https://www.cdc.gov/poxvirus/mpox/response/2022/us-map.html>.
- (10) Waddell, C. J.; Filardo, T. D.; Prasad, N.; Pellegrini, G. J.; Persad, N.; Carson, W. C.; Navarra, T.; Townsend, M. B.; Satheshkumar, P. S.; Lowe, D.; Borne, D.; Janssen, J.; Okoye, N.; Bejarano, A.; Marx, G. E.; Mosites, E. Possible Undetected Mpox Infection Among Persons Accessing Homeless Services and Staying in Encampments—San Francisco, California, October–November 2022. *Morb. Mortal. Wkly. Rep.* **2023**, *72* (9), 227–231.
- (11) Peccia, J.; Zulli, A.; Brackney, D. E.; Grubaugh, N. D.; Kaplan, E. H.; Casanovas-Massana, A.; Ko, A. I.; Malik, A. A.; Wang, D.; Wang, M.; Warren, J. L.; Weinberger, D. M.; Arnold, W.; Omer, S. B. Measurement of SARS-CoV-2 RNA in Wastewater Tracks Community Infection Dynamics. *Nat. Biotechnol.* **2020**, *38* (10), 1164–1167.
- (12) Zulli, A.; Pan, A.; Bart, S. M.; Crawford, F. W.; Kaplan, E. H.; Cartter, M.; Ko, A. I.; Sanchez, M.; Brown, C.; Cozens, D.; Brackney, D. E.; Peccia, J. Predicting Daily COVID-19 Case Rates from SARS-CoV-2 RNA Concentrations across a Diversity of Wastewater Catchments. *FEMS Microbes* **2022**, *2*, No. xtab022.
- (13) Wolfe, M. K.; Yu, A. T.; Duong, D.; Rane, M. S.; Hughes, B.; Chan-Herur, V.; Donnelly, M.; Chai, S.; White, B. J.; Vugia, D. J.; Boehm, A. B. Use of Wastewater for Mpox Outbreak Surveillance in California. *N. Engl. J. Med.* **2023**, *388* (6), 570–572.
- (14) Boehm, A. B.; Hughes, B.; Duong, D.; Chan-Herur, V.; Buchman, A.; Wolfe, M. K.; White, B. J. Wastewater Concentrations of Human Influenza, Metapneumovirus, Parainfluenza, Respiratory Syncytial Virus, Rhinovirus, and Seasonal Coronavirus Nucleic-Acids during the COVID-19 Pandemic: A Surveillance Study. *Lancet Microbe* **2023**, *4* (5), e340–e348, DOI: 10.1016/S2666-5247(22)00386-X.
- (15) Guo, Y.; Li, J.; O'Brien, J.; Sivakumar, M.; Jiang, G. Back-Estimation of Norovirus Infections through Wastewater-Based Epidemiology: A Systematic Review and Parameter Sensitivity. *Water Res.* **2022**, *219*, No. 118610.

- (16) Carneiro, H. A.; Mylonakis, E. Google Trends: A Web-Based Tool for Real-Time Surveillance of Disease Outbreaks. *Clin. Infect. Dis.* **2009**, *49* (10), 1557–1564.
- (17) Cervellin, G.; Comelli, L.; Lippi, G. Is Google Trends a Reliable Tool for Digital Epidemiology? Insights from Different Clinical Settings. *J. Epidemiol. Glob. Health* **2017**, *7* (3), 185–189.
- (18) Yuan, K.; Huang, G.; Wang, L.; Wang, T.; Liu, W.; Jiang, H.; Yang, A. C. Predicting Norovirus in the United States Using Google Trends: Infodemiology Study. *J. Med. Internet Res.* **2021**, *23* (9), No. e24554.
- (19) Samaras, L.; García-Barriocanal, E.; Sicilia, M.-A. Syndromic Surveillance Models Using Web Data: The Case of Influenza in Greece and Italy Using Google Trends. *JMIR Public Health Surveill.* **2017**, *3* (4), No. e8015.
- (20) Butler, D. When Google Got Flu Wrong. *Nature* **2013**, *494* (7436), 155–156.
- (21) Lazer, D.; Kennedy, R.; King, G.; Vespignani, A. The Parable of Google Flu: Traps in Big Data Analysis. *Science* **2014**, *343* (6176), 1203–1205.
- (22) Morsy, S.; Dang, T. N.; Kamel, M. G.; Zayan, A. H.; Makram, O. M.; Elhady, M.; Hirayama, K.; Huy, N. T. Prediction of Zika-Confirmed Cases in Brazil and Colombia Using Google Trends. *Epidemiol. Infect.* **2018**, *146* (13), 1625–1627.
- (23) Santangelo, O. E.; Provenzano, S.; Piazza, D.; Giordano, D.; Calamusa, G.; Firenze, A. Digital Epidemiology: Assessment of Measles Infection through Google Trends Mechanism in Italy. *Ann. Ig.* **2019**, *31*, 385–391.
- (24) Szmuda, T.; Ali, S.; Hetzger, T. V.; Rosvall, P.; Sloniewski, P. Are Online Searches for the Novel Coronavirus (COVID-19) Related to Media or Epidemiology? A Cross-Sectional Study. *Int. J. Infect. Dis.* **2020**, *97*, 386–390.
- (25) Cho, S.; Sohn, C. H.; Jo, M. W.; Shin, S.-Y.; Lee, J. H.; Ryoo, S. M.; Kim, W. Y.; Seo, D.-W. Correlation between National Influenza Surveillance Data and Google Trends in South Korea. *PLoS One* **2013**, *8* (12), No. e81422.
- (26) Nuti, S. V.; Wayda, B.; Ranasinghe, I.; Wang, S.; Dreyer, R. P.; Chen, S. I.; Murugiah, K. The Use of Google Trends in Health Care Research: A Systematic Review. *PLoS One* **2014**, *9* (10), No. e109583.
- (27) Gluskin, R. T.; Johansson, M. A.; Santillana, M.; Brownstein, J. S. Evaluation of Internet-Based Dengue Query Data: Google Dengue Trends. *PLoS Neglected Trop. Dis.* **2014**, *8* (2), No. e2713.
- (28) Althouse, B. M.; Ng, Y. Y.; Cummings, D. A. T. Prediction of Dengue Incidence Using Search Query Surveillance. *PLoS Neglected Trop. Dis.* **2011**, *5* (8), No. e1258.
- (29) Chan, E. H.; Sahai, V.; Conrad, C.; Brownstein, J. S. Using Web Search Query Data to Monitor Dengue Epidemics: A New Model for Neglected Tropical Disease Surveillance. *PLoS Neglected Trop. Dis.* **2011**, *5* (5), No. e1206.
- (30) Boehm, A. B.; Shelden, B.; Duong, D.; Banaei, N.; White, B. J.; Wolfe, M. K. A Retrospective Longitudinal Study of Adenovirus Group F, Norovirus GI and GII, Rotavirus, and Enterovirus Nucleic Acids in Wastewater Solids at Two Wastewater Treatment Plants: Solid-Liquid Partitioning and Relation to Clinical Testing Data. *mSphere* **2024**, *9* (3), No. e00736-23.
- (31) Roldan-Hernandez, L.; Boehm, A. B. Adsorption of Respiratory Syncytial Virus, Rhinovirus, SARS-CoV-2, and F+ Bacteriophage MS2 RNA onto Wastewater Solids from Raw Wastewater. *Environ. Sci. Technol.* **2023**, *57* (36), 13346–13355.
- (32) Roldan-Hernandez, L.; Oost, C. V.; Boehm, A. Solid-Liquid Partitioning of Dengue, West Nile, Zika, Hepatitis A, Influenza A, and SARS-CoV-2 Viruses in Wastewater from across the USA. *Environ. Sci.: Water Res. Technol.* **2024**, *11*, 88–99, DOI: 10.1039/D4EW00225C.
- (33) Jothikumar, N.; Lowther, J. A.; Henshilwood, K.; Lees, D. N.; Hill, V. R.; Vinjé, J. Rapid and Sensitive Detection of Noroviruses by Using TaqMan-Based One-Step Reverse Transcription-PCR Assays and Application to Naturally Contaminated Shellfish Samples. *Appl. Environ. Microbiol.* **2005**, *71* (4), 1870–1875.
- (34) Li, Y.; Zhao, H.; Wilkins, K.; Hughes, C.; Damon, I. K. Real-Time PCR Assays for the Specific Detection of Monkeypox Virus West African and Congo Basin Strain DNA. *J. Virol. Methods* **2010**, *169* (1), 223–227.
- (35) Foodborne Outbreak Investigations in Connecticut. CT.gov—Connecticut's Official State Website, 2023. <https://portal.ct.gov/DPH/Epidemiology-and-Emerging-Infections/Foodborne-Outbreak-Investigations-in-Connecticut>.
- (36) FAQ about Google Trends data—Trends Help, 2023. https://support.google.com/trends/answer/4365533?hl=en&ref_topic=6248052&sjid=12390365559024080290-NA.
- (37) Create Daily Series from Google Trends, 2023. <https://trendecon.github.io/trendecon/>.
- (38) Eichenauer, V. Z.; Indergand, R.; Martínez, I. Z.; Sax, C. Obtaining Consistent Time Series from Google Trends. *Econ. Inq.* **2022**, *60* (2), 694–705.
- (39) Kaplan, E. H.; Zulli, A.; Sanchez, M.; Peccia, J. Scaling SARS-CoV-2 Wastewater Concentrations to Population Estimates of Infection. *Sci. Rep.* **2022**, *12* (1), No. 3487.
- (40) McMahan, C. S.; Self, S.; Rennert, L.; Kalbaugh, C.; Kriebel, D.; Graves, D.; Colby, C.; Deaver, J. A.; Popat, S. C.; Karanfil, T.; Freedman, D. L. COVID-19 Wastewater Epidemiology: A Model to Estimate Infected Populations. *Lancet Planet. Health* **2021**, *5* (12), e874–e881.
- (41) Brouwer, A. F.; Eisenberg, J. N. S.; Pomeroy, C. D.; Shulman, L. M.; Hindiyeh, M.; Manor, Y.; Grotto, I.; Koopman, J. S.; Eisenberg, M. C. Epidemiology of the Silent Polio Outbreak in Rahat, Israel, Based on Modeling of Environmental Surveillance Data. *Proc. Natl. Acad. Sci. U.S.A.* **2018**, *115* (45), E10625–E10633.
- (42) Norovirus National Trends—NREVSS | CDC, 2023. <https://www.cdc.gov/surveillance/nrevss/norovirus/natl-trend.html>.
- (43) Oren, E.; Frere, J.; Yom-Tov, E.; Yom-Tov, E. Respiratory Syncytial Virus Tracking Using Internet Search Engine Data. *BMC Public Health* **2018**, *18* (1), 445.
- (44) Crowson, M. G.; Witsell, D.; Eskander, A. Using Google Trends to Predict Pediatric Respiratory Syncytial Virus Encounters at a Major Health Care System. *J. Med. Syst.* **2020**, *44* (3), 57.
- (45) Wang, D.; Guerra, A.; Wittke, F.; Lang, J. C.; Bakker, K.; Lee, A. W.; Finelli, L.; Chen, Y.-H. Real-Time Monitoring of Infectious Disease Outbreaks with a Combination of Google Trends Search Results and the Moving Epidemic Method: A Respiratory Syncytial Virus Case Study. *Trop. Med. Infect. Dis.* **2023**, *8* (2), 75.
- (46) Bhatt, P.; Parmar, N.; Parekh, T.; Pemmasani, S.; Shaikh, N.; Boateng, A. J.; Donda, K.; Doshi, H.; Dapaah-Siakwan, F. Predicting Lead-Time RSV-Related Pediatric Hospitalizations From Historic Google Trend Search. *Hosp. Pediatr.* **2023**, *13* (11), e325–e328.
- (47) Osuka, H.; Hall, A. J.; Wikswo, M. E.; Baker, J. M.; Lopman, B. A. Temporal Relationship Between Healthcare-Associated and Nonhealthcare-Associated Norovirus Outbreaks and Google Trends Data in the United States. *Infect. Control Hosp. Epidemiol.* **2018**, *39* (3), 355–358.
- (48) Desai, R.; Hall, A. J.; Lopman, B. A.; Shimshoni, Y.; Rennick, M.; Efron, N.; Matias, Y.; Patel, M. M.; Parashar, U. D. Norovirus Disease Surveillance Using Google Internet Query Share Data. *Clin. Infect. Dis.* **2012**, *55* (8), e75–e78.
- (49) Ondrikova, N.; Harris, J. P.; Douglas, A.; Hughes, H. E.; Iturriza-Gomara, M.; Vivancos, R.; Elliot, A. J.; Cunliffe, N. A.; Clough, H. E. Predicting Norovirus in England Using Existing and Emerging Syndromic Data: Infodemiology Study. *J. Med. Internet Res.* **2023**, *25* (1), No. e37540.
- (50) Akram, H. Assessing Google Search Trends for Mpox in the United States. *Apollo Med.* **2024**, *21* (1), S83–S87.
- (51) Comeau, N.; Abdelnour, A.; Ashack, K. Assessing Public Interest in Mpox via Google Trends, YouTube, and TikTok. *JMIR Dermatol.* **2023**, *6*, No. e48827.
- (52) Blum, D. *Oh, No. Time to Worry About Norovirus Again*; The New York Times, 2023.
- (53) Flaxman, S.; Mishra, S.; Gandy, A.; Unwin, H. J. T.; Mellan, T. A.; Coupland, H.; Whittaker, C.; Zhu, H.; Berah, T.; Eaton, J. W.;

Monod, M.; Ghani, A. C.; Donnelly, C. A.; Riley, S.; Vollmer, M. A. C.; Ferguson, N. M.; Okell, L. C.; Bhatt, S.; et al. Estimating the Effects of Non-Pharmaceutical Interventions on COVID-19 in Europe. *Nature* **2020**, *584* (7820), 257–261.

(54) Olson, D. R.; Konty, K. J.; Paladini, M.; Viboud, C.; Simonsen, L. Reassessing Google Flu Trends Data for Detection of Seasonal and Pandemic Influenza: A Comparative Epidemiological Study at Three Geographic Scales. *PLOS Comput. Biol.* **2013**, *9* (10), No. e1003256.



CAS BIOFINDER DISCOVERY PLATFORM™

PRECISION DATA FOR FASTER DRUG DISCOVERY

CAS BioFinder helps you identify
targets, biomarkers, and pathways

Unlock insights

CAS
A division of the
American Chemical Society